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Interference analysis between NGSO systems based on the beam coverage and link angle

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Abstract. The existing interference assessment methods usually consider the special scenario of earth station coincidence. However, in practice, earth stations such as gate stations are sparsely distributed and often do not coincide. Aiming at the problem of how to efficiently calculate the co-frequency interference between Non-Geostationary Satellite Orbit (NGSO) systems with high complexity and a large amount of data, the influencing factors of interference scenes between NGSO systems with non-earth station coincidence are analyzed. The time-varying expression of the bit error rate (BER) of the interference evaluation index to measure the system performance is derived. An interference analysis method based on the beam coverage and link angle is proposed. In this method, four interference time-varying parameters are reduced to two. One of them is associated with the distance between earth stations, which is used to rapidly evaluate the probability of harmful interference between NGSO systems. It provides a reference for the construction of the spacing of earth stations.

1. Introduction

In recent years, the explosive growth of information and data has made the demand for broadband satellite network communication increasingly urgent, and spatial spectrum and orbit resources become scarce. With the increasing congestion of geostationary orbit (GSO), the constellation system of non-geostationary orbit has made rapid development. At present, commercial space companies represented by SpaceX and OneWeb have launched low earth orbit (LEO) satellites operating in Ku/Ka band. Spectrum sharing among multiple NGSO constellation systems is one of the key technologies to improve frequency utilization, but it also inevitably leads to potential co-frequency interference between NGSO satellites. Therefore, the large-scale interference analysis method between NGSO systems has become an indispensable content in the construction of NGSO constellation systems^[1].

The coexisting scene of the NGSO constellation system is complex, and the interference calculation data is large, so the public research is relatively few. Nelson and Pritchard proposed the basic orbit and interference theory between LEO satellites in the NGSO constellation system^[2]. Jin et al. proposed a link angle analysis method for the specific scenario of the co-frequency interference with the coincidence of earth stations between NGSO systems. On this basis, they applied it to the global distributed earth station interference analysis scenario and proposed the probability calculation method of harmful interference between NGSO systems^[3]. According to the multi-beam and dynamic characteristics of the NGSO satellite, Zhao calculated and evaluated the interference between different NGSO constellations through the co-simulation platform and explored the influence of modeling



parameters such as simulation step size, isolation distance, and latitude position of ground stations on the simulation results^[4]. The existing interference assessment methods usually consider the special scenario of earth station coincidence. When considering the distance between earth stations, the dynamic time-varying parameters of the interference scene increase. The scene is complex, which brings greater challenges for the rapid interference assessment method. In addition, the above interference assessment methods all use the ITU-recommended dry-to-noise ratio, which cannot reflect the performance of the interfered NGSO constellation system. How to select suitable interference evaluation indexes to efficiently evaluate the co-frequency interference with high complexity and a large amount of interference calculation data is a difficult problem.

In this paper, a basic link budget model for interference analysis between LEO satellites is established. The range of interference time-varying parameters is determined by joint beam coverage so as to establish the constraint conditions between the BER threshold and the key influencing factors of interference. Secondly, the statistical analysis algorithm flow of the occurrence probability of harmful interference under different spacing of earth stations is given to evaluate the overall disturbance of the NGSO constellation system. Finally, taking the large-scale Starlink system^[5] and the OneWeb system^[6] as examples, the effectiveness of the proposed method for the co-frequency interference analysis is verified. At the same time, the influence of earth station spacing on the occurrence probability of harmful interference in the NGSO constellation system is simulated, which provides a reference for interference avoidance between NGSO constellation systems.

2. Calculation of co-frequency interference between NGSO satellites

The upstream and downstream link interference scenes between NGSO satellites are similar. This paper takes the upstream link interference as an example to analyze the interference. The uplink interference scene between NGSO satellites is shown in Figure 1.

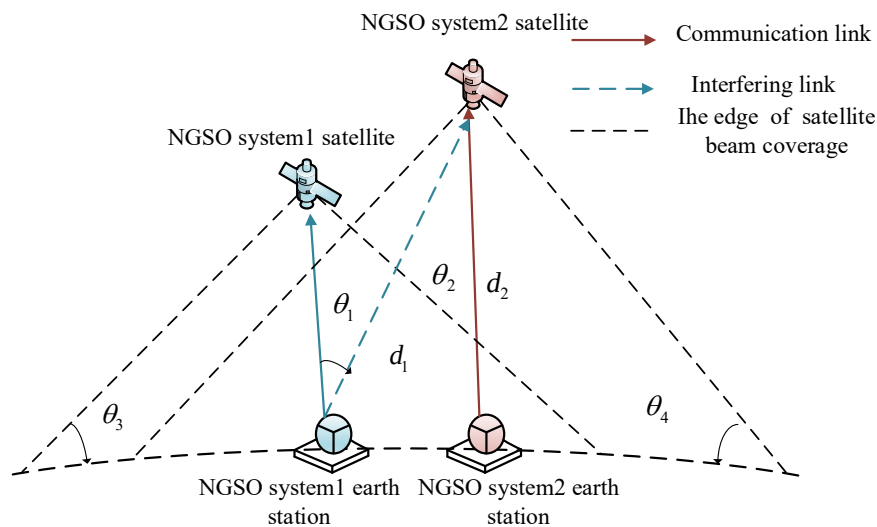


Figure 1. Uplink interference scenario between NGSO satellites

As shown in Figure 1, the NGSO system 2 satellite receives useful signals from the NGSO system 2 earth station while receiving interference signals from the NGSO system 1 earth station. θ_1 is the included angle of the interference link. θ_2 is the included angle of the communication link. θ_3 is the satellite beam coverage edge angle of NGSO system 1, θ_4 is the satellite beam coverage edge angle of NGSO system 2. d_1 and d_2 respectively, represent the interference link distance and the communication link distance^[7].

The carrier-to-interference plus noise ratio can be expressed as:

$$\frac{C}{I + N} = \frac{P_{1,T} G_{1,T}(0) G_{1,R}(0) (\lambda_1 / 4\pi d_2)^2}{P'_{sat,T,1} G_{sat,T,1}(\theta_1) G_{cs,R,2}(\theta_2) (\lambda / 4\pi d_1)^2 + kTB_{ng2}} \quad (1)$$

where $P_{1,T}$ is the transmit power of the communication signal, $G_{1,T}(0)$ is the transmit antenna maximum gain of NGSO system 2 earth station, and $G_{1,R}(0)$ is receive antenna maximum gain of the NGSO system 2 satellite. $P'_{sat,T,1}$ is the in-band power of the interference signal, $G_{sat,T,1}(\theta_1)$ is the transmit gain corresponding to the θ_1 angle of the NGSO system 1 earth station deviating from the interference link, and $G_{cs,R,2}(\theta_2)$ is the receive gain corresponding to the θ_2 angle of NGSO system 2 satellites deviating from the interference link. λ and λ_1 respectively, represent the wavelength of the interference signal and communication signal. k is the Boltzmann constant, T and B_{ng2} respectively, represent the equivalent noise temperature and the communication bandwidth of the receive.

For the determined NGSO system, BER can be used as the interference evaluation index to further evaluate the system's performance. Taking QPSK modulation commonly used in the NGSO constellation system as an example, the relationship between the BER of communication at the receiving end of the system and the normalized signal-to-noise ratio under this modulation is:

$$P_e = \frac{1}{2} \left(1 - \operatorname{erf} \sqrt{\frac{C \cdot B_{ng2}}{(N + I) \cdot R}} \right) \quad (2)$$

where erf is the Gaussian error function, E_b is 1-bit energy, n_0 is normalized noise, and R is data rate.

In the interference scenario shown in Figure 1, the angle of the interference link, the angle of the communication link, the distance of the interference link, and the distance of the communication link all change dynamically during the operation of the NGSO satellite. The time-varying parameters mainly include θ_1 , θ_2 , d_1 and d_2 . Therefore, in order to accurately calculate the interference among NGSO constellation systems, it is necessary to calculate the interference evaluation index under the dynamic changes of multiple parameters, which has a huge amount of calculated data. In order to quickly evaluate the interference, it is necessary to further consider the range of time-varying parameters and find out the key influencing factors.

In the actual working scenario, the spaceborne beam range of NGSO satellites is limited. In this case, the time-varying parameters θ_1 , θ_2 , d_1 and d_2 have a limited range of variation. The edge angle of the satellite beam coverage of NGSO system 1 is approximately the lowest communication elevation Angle $eth1$, and the edge Angle of satellite beam coverage of NGSO system 2 is approximately the lowest communication elevation Angle $eth2$. Under the constraint of spaceborne multi-beam coverage, the propagation distance of the NGSO system 2 satellite beam covering edge link is:

$$d_{\max} = \sqrt{(R_e + h_2)^2 - R_e^2 \cos^2 eth2} - R_e \sin eth2 \quad (3)$$

According to the spatial geometric relationship in the scene in Figure 1, the variation range of the time-varying BER affecting parameters θ_1 , θ_2 , d_1 and d_2 are as follows.

$$\Delta\theta_1 = \pi - eth1 - eth2 \quad (4)$$

$$\Delta\theta_2 = 2 \arccos \frac{d_{\max}^2 + (R_e + h_2)^2 - R_e^2}{2(R_e + h_2)d_{\max}} \quad (5)$$

$$\Delta d_1 = \Delta d_2 = d_{\max} - h_2 \quad (6)$$

When the BER is less than 0.05, the satellite communication uplink is not interfered [8]. Therefore, the bit error rate of 0.05 can be taken as the threshold of the co-frequency interference. By analyzing the key influencing factors θ_1 , θ_2 and their influence on the BER, the satellite co-frequency can be transformed so as to quickly evaluate the interference between NGSO constellation systems.

3. Statistical analysis of the occurrence probability of harmful interference between NGSO systems

Generally, in order to evaluate the co-frequency interference among large-scale NGSO constellation systems, the occurrence probability of harmful interference is selected as the evaluation index of harmful interference between systems.

The influence of the distance between the earth stations on the link angle θ_2 is further analyzed. The minimum value of θ_2 is obtained when the interfered NGSO system 2 satellite moves to the minimum communication elevation angle of the NGSO system 2 earth station. The maximum value of θ_2 is obtained when the NGSO system 2 satellite moves above the middle of the two earth stations. Therefore, the variation range of θ_2 is very small. The case with the worst interference is considered here; that is, the minimum value of θ_2 is taken.

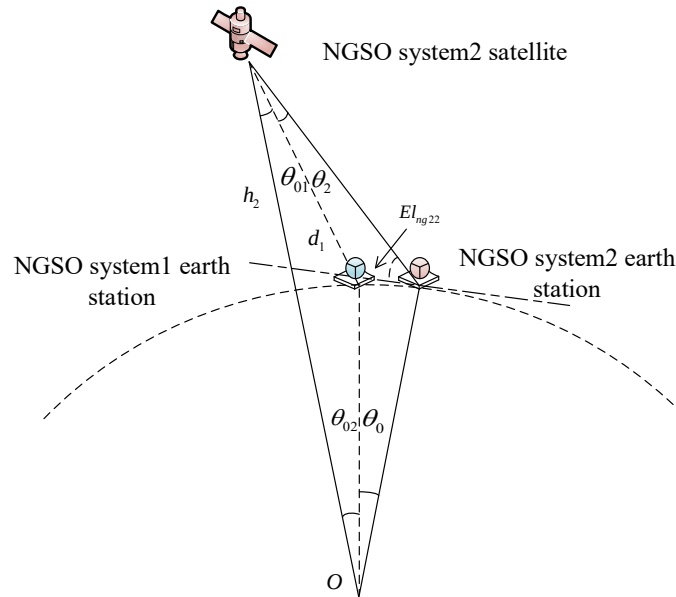


Figure 2. Spatial relationship of NGSO system 2 satellite located at the minimum communication elevation Angle of NGSO system 2 earth station

As shown in the Figure 2, it can be expressed as:

$$\theta_{02} = 180^\circ - (El_{ng22} + 90^\circ) - \frac{180 \cdot d}{\pi R_e} - \arcsin \frac{R_e \cdot \sin(El_{ng22} + 90^\circ)}{h_2 + R_e} \quad (7)$$

where El_{ng22} is the elevation angle of NGSO system 2 satellite, R_e is the radius of the earth, and d is the distance between earth stations.

According to the law of cosine, the distance between the earth stations can be expressed as:

$$d_1 = \sqrt{(h_2 + R_e)^2 + R_e^2 - 2(h_2 + R_e)R_e \cos \theta_{02}} \quad (8)$$

where θ_{02} is the included Angle formed by the line between the earth station of NGSO system 1, the line between the satellite of NGSO system 2, and the earth center.

θ_2 can be expressed as:

$$\theta_2 = \arcsin \frac{R_e \cdot \sin(El_{ng22} + 90^\circ)}{h_2 + R_e} - \arcsin \frac{R_e \cdot \sin \theta_{02}}{d_1}. \quad (9)$$

If the distance between the earth stations is determined, the corresponding value is determined.

According to the proposed interference analysis method based on the beam coverage and link angle, the threshold corresponding to the BER reaching the threshold of 0.05 can be calculated according to the determined value. As long as the occurrence probability statistical analysis of a variable θ_1 is carried out, the number of combinations exceeding the system tolerance can be obtained. The occurrence probability of harmful interference under the deployment spacing of the earth station can be further obtained.

NGSO system 1 is assumed to have a total of M NGSO satellites, denoted as set $S_1 = \{s_1^1, s_1^2, \dots, s_1^M\}$, and NGSO system 2 has a total of N NGSO satellites, denoted as set $S_2 = \{s_2^1, s_2^2, \dots, s_2^N\}$. When considering the deployment spacing of earth stations, the statistical analysis process of the occurrence probability of harmful interference between systems is as follows

Step 1: Based on constellation configurations of different NGSO systems 1 and 2, real-time position information of NGSO satellites in the constellation is obtained.

Step 2: As the earth is approximately spherical in distribution, the geographical latitude is 1° apart from the same line, so the actual geographical distance is about 111 km. Therefore, different latitude difference values are set to represent the distance between the earth stations.

Step 3: Based on the interference analysis method based on the beam coverage and link Angle, determine the threshold values θ_{1th} corresponding to different earth station spacing θ_2 .

Step 4: For the k simulation moment, the visual relationship between satellite and earth station P_1 and P_2 in S_1 and S_2 are cycled, respectively. The visual satellite sets $S_{1i} = \{s_{1i}^1, s_{1i}^2, \dots, s_{1i}^{M'}\}$ and $S_{2i} = \{s_{2i}^1, s_{2i}^2, \dots, s_{2i}^{N'}\}$ of the i earth station P_{1i} and P_{2i} are obtained, with the number of elements being M_i' and N_i' respectively. It is assumed that the earth station can access set $S_{1i}' = \emptyset$.

Step 5: We obtain the element s_{1i}^h ($1 \leq h < M'$) in the visual satellite set S_{1i}' of NGSO system 1, and the link angle $\theta_{h,j}$ between the earth station P_{1i} and element s_{2i}^j ($1 \leq j \leq N'$) in the visual satellite set S_{2i}' .

Step 6: For all $\theta_{h,j}$ corresponding to s_{1i}^h in Step 5, if $\exists \theta_{h,j} \leq \theta_{1th}$, then $s_{1i}^h \in S_{1i}'$, otherwise $s_{1i}^h \notin S_{1i}'$, S_{1i}' element number is denoted as V_i .

Step 7: Steps 5 to 6 are repeated until all elements in S_{1i}' are traversed.

Step 8: Steps 4 to 7 are repeated until all L earth stations P_{1i} have completed traversal.

Step 9: At the moment k, the occurrence probability of harmful interference on the receiving end of the satellite of NGSO system 2 is $\eta_k = 1 / L \sum_{i=1}^L V_i / (M_i' * N_i')$, $k = k + 1$, $\eta_k \in (0, 1)$.

Step 10: Steps 2 to 9 are repeated until all k simulation moments are traversed.

Under the constraints of beam coverage, the combination ratio of the link angle within the threshold angle is obtained. The occurrence probability of harmful interference occurring in a certain time is obtained. The higher the occurrence probability of harmful interference is, the lower the availability of the system is, so as to measure the frequency compatibility between NGSO constellation systems.

4. Simulation analysis

This section aims to solve the proposed interference analysis method and explore the effect of earth station spacing on the occurrence probability of harmful interference between NGSO systems.

4.1. Simulation parameters

The link budget parameters of the two systems are determined as shown in Table 1. Starlink system is used as the interfering system while OneWeb system is used as the interfered system.

Table 1. The link budget parameters of Oneweb and Starlink systems.

Parameter	Value of Oneweb system	Value of Starlink system
Orbit height	1200 km	550 km
Transmit power of earth station	21.7 dBW	40 dBW
Earth station antenna type	AP8	AP8
Constellation configuration	18×40	22×72
Maximum gain of earth station antenna	21 dBi	48 dBi
Orbit inclination	87.9°	53°
Maximum gain of satellite antenna	31.2 dBi	41.0 dBi
Satellite antenna type	ITU.R S1528	ITU.R S1528
System noise temperature	600 K	600 K
Bandwidth	250 MHz	500 MHz
Carrier frequency	27.725 GHz	27.725 GHz

4.2. Simulation analysis of co-frequency interference between NGSO satellites

Under the above simulation parameters, the variation of BER at the receiving end of OneWeb satellite with the included angle of interference link and communication link is analyzed. This is used to verify the validity of the co-frequency interference analysis.

According to Formulas (4)-(6), the shortest link distance of free space attenuation of the interfering signal sent by the earth station of Starlink constellation system within the coverage range of the interfered satellite beam of OneWeb constellation system is 1200 km, and the longest link distance is the covering edge link propagation distance^[9], which is about 1412 km. That is, the variation range of d_1 and d_2 are 1200 km~1412 km, the variation range of θ_1 is 0°~85°, and the variation range of θ_2 is 0°~58°.

According to Formulas (7)-(9), the latitude difference of the earth station is used to represent the spacing of different earth stations. The spacing of the earth stations is selected as 0 km, 55 km, 111 km, 222 km, 333 km, 444 km, and 555 km. In the worst case, 0°, 2°, 4°, 8°, 13°, 18°, and 23° are approximately taken as examples to analyze the system level interference related to the angle of communication link and the distance between earth stations.

In order to directly reflect the influence of interference time-varying parameters θ_1 , θ_2 , d_1 and d_2 on the communication BER, the BER fluctuation difference caused by them is statistically analyzed. The maximum fluctuation difference caused by interference link distance and

communication link distance is 0.0127. The maximum fluctuation difference caused by the interference link angle and communication link angle is 0.4895 and 0.4391, respectively.

In order to quickly evaluate the occurrence probability of harmful interference between NGSO satellites in the future, it is necessary to analyze the change of BER with the corresponding value and determine the value under the set BER threshold when it is associated with the distance between earth stations. Figure 3 shows the different approximate values corresponding to the worst case of the earth station spacing of 0 km, 55 km, 111 km, 222 km, 333 km, 444 km, and 555 km, as well as the variation of BER with the angle of interference link between NGSO satellites when the maximum value is 58° .

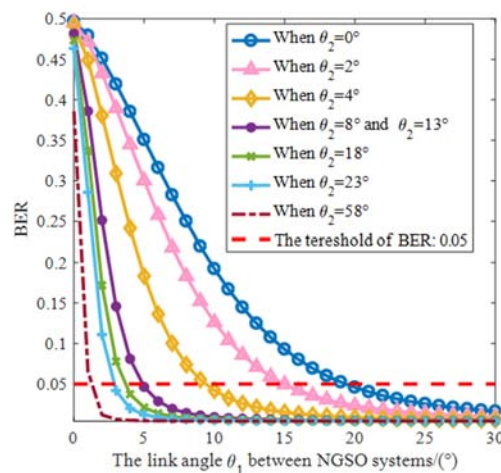


Figure 3. Graph of bit error rate variation with link angle between NGSO satellites

4.3. Simulation analysis of occurrence probability of harmful interference between NGSO systems

In this section, the influence of the occurrence probability of harmful interference under different distances between earth stations is simulated and analyzed. The simulation cycle is 120 minutes in total. In the simulation, the latitude difference of the longitude, latitude, and altitude of the earth station in the earth-centered fixed coordinate system is set to represent different earth station spacing.

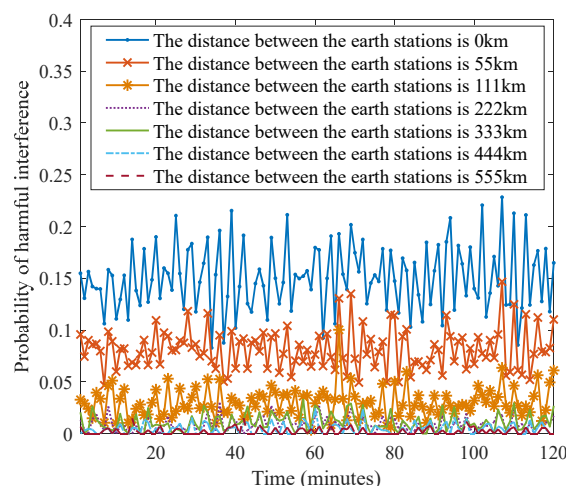


Figure 4. Occurrence probability of harmful interference varies with the distance between earth stations

The simulation results in Figure 4 show that collinear interference is inevitable when the signal gate stations coincide. The Starlink system has the greatest influence on the occurrence probability of

harmful interference between OneWeb systems. In the worst case, the occurrence probability of harmful interference is 22.83%, and the occurrence probability of harmful interference changes obviously when the distance between earth stations varies from 0 to 111 km. To some extent, increasing the distance between earth stations can reduce the occurrence probability of harmful interference, and interference coordination can be carried out by planning the location of gateway stations and user stations, providing a reference for the construction of large-scale NGSO systems in the future.

5. Conclusion

This paper studies the problem of the co-frequency interference between large-scale NGSO constellation systems. The interference analysis method based on beam coverage and link angle is used to carry out the mathematical modeling of the co-frequency interference scene between NGSO systems. The calculation method of BER in the dynamic scene is given, as well as the variation range of the parameters of the time varying BER. Therefore, the change of communication BER caused by the co-frequency interference can be converted into link angle constraint, which lays a foundation for the rapid evaluation of the interference between NGSO systems. In addition, based on the interference analysis between NGSO satellites, the influence of the deployment spacing of earth stations on the frequency compatibility between large-scale NGSO constellations is studied.

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